

Fig. 2.5

Thus, for any two rational numbers, their average/mid point is rational. Proceeding similarly, we can generate infinitely many rational numbers.

Example 2.1 Find any two rational numbers between $\frac{1}{2}$ and $\frac{2}{3}$.

Solution 1

A rational number between $\frac{1}{2}$ and $\frac{2}{3} = \frac{1}{2}\left(\frac{1}{2}+\frac{2}{3}\right)=\frac{1}{2}\left(\frac{3+4}{6}\right)=\frac{1}{2}\left(\frac{7}{6}\right)=\frac{7}{12}$

A rational number between $\frac{1}{2}$ and $\frac{7}{12} = \frac{1}{2}\left(\frac{1}{2}+\frac{7}{12}\right)=\frac{1}{2}\left(\frac{6+7}{12}\right)=\frac{1}{2}\left(\frac{13}{12}\right)=\frac{13}{24}$

Hence two rational numbers between $\frac{1}{2}$ and $\frac{2}{3}$ are $\frac{7}{12}$ and $\frac{13}{24}$ (of course, there are many more!)

There is an interesting result that could help you to write instantly rational numbers between any two given rational numbers.

Result

If $\frac{p}{q}$ and $\frac{r}{s}$ are any two rational numbers such that $\frac{p}{q}<\frac{r}{s}$, then $\frac{p+r}{q+s}$ is a rational number, such that $\frac{p}{q}<\frac{p+r}{q+s}<\frac{r}{s}$.

Let us take the same example: Find any two rational numbers between $\frac{1}{2}$ and $\frac{2}{3}$

Solution 2

$\frac{1}{2}<\frac{2}{3}$ gives $\frac{1}{2}<\frac{1+2}{2+3}<\frac{2}{3}$ *or* $\frac{1}{2}<\frac{3}{5}<\frac{2}{3}$ gives $\frac{1}{2}<\frac{1+3}{2+5}<\frac{3}{5}<\frac{3+2}{5+3}<\frac{2}{3}$ *or* $\frac{1}{2}<\frac{4}{7}<\frac{3}{5}<\frac{5}{8}<\frac{2}{3}$

Solution 3

Any more new methods to solve? Yes, if decimals are your favourites, then the above example can be given an alternate solution as follows:

$\frac{1}{2} = 0.5$ and $\frac{2}{3} = 0.66...$

Hence rational numbers between $\frac{1}{2}$ and $\frac{2}{3}$ can be listed as 0.51, 0.57,0.58,…

Solution 4

There is one more way to solve some problems. For example, to find four rational numbers between $\frac{4}{9}$ and $\frac{3}{5}$, note that the LCM of 9 and 5 is 45; so we can write $\frac{4}{9} = \frac{20}{45}$ and $\frac{3}{5} = \frac{27}{45}$.

Therefore, four rational numbers between $\frac{4}{9}$ and $\frac{3}{5}$ are $\frac{21}{45}$, $\frac{22}{45}$, $\frac{23}{45}$, $\frac{24}{45}$, ...

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